

CHAPTER 19

CHEMICAL COORDINATION AND INTEGRATION

KEY CONCEPT

- Body functions are regulated and coordinated by chemical messengers and nervous system.
- Endocrine system uses hormones, while nervous system uses electrical impulses.

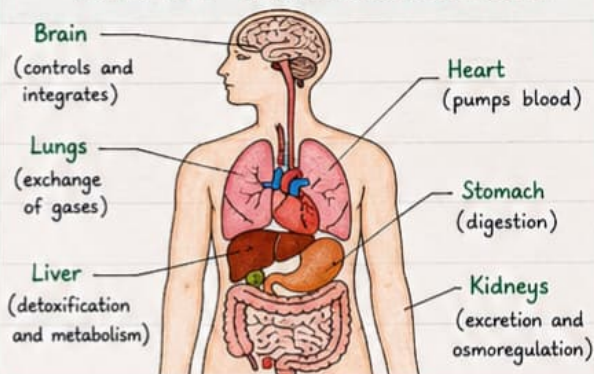
Coordination allows living organisms to respond to changes in their internal and external environment and maintain a stable internal environment.

1 NEED FOR COORDINATION

- Multicellular organisms have millions of cells performing specialised functions.
- Changes in environment occur continuously.
- Coordination ensures that all parts of the body work together for survival.

Example : When we touch a hot object, our body quickly responds to withdraw the hand and avoid injury.

HUMAN BODY – A WELL COORDINATED SYSTEM



2 MODES OF CHEMICAL COORDINATION

There are two main modes :

(A) ENDOCRINE SYSTEM (Hormonal coordination)

→ Involves secretion of hormones by endocrine glands which act on specific target organs.

(B) NERVOUS SYSTEM (Neural coordination)

→ Involves electrical impulses transmitted by neurons for a very fast response.

ENDOCRINE SYSTEM

Uses chemical messengers (hormones)

Slower response

Long-lasting effects

Acts on specific target organs via blood

NERVOUS SYSTEM

Uses electrical impulses (neurotransmitters)

Very fast response

Short-lived effects

Acts on effectors (muscles/glands) via neurons

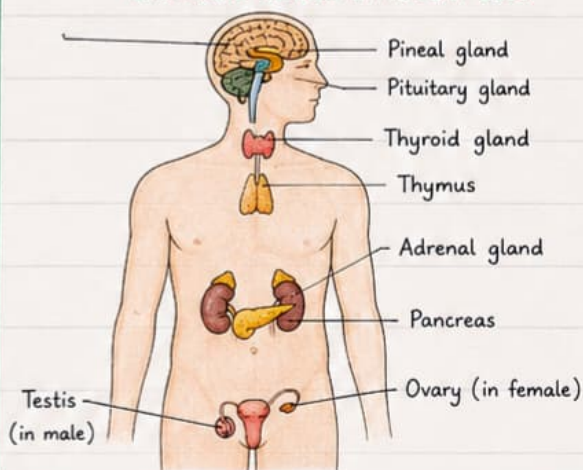
3 ENDOCRINE SYSTEM

- Consists of ductless glands that secrete hormones directly into the bloodstream.

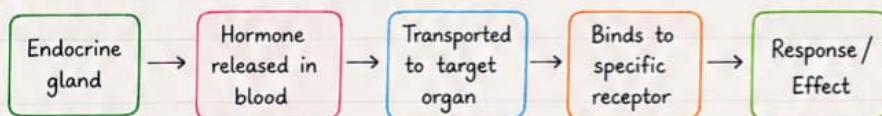
MAJOR ENDOCRINE GLANDS

- ♥ Pituitary gland – Master gland
- ♥ Thyroid gland – Regulates metabolism
- ♥ Parathyroid glands – Regulate calcium and phosphorus
- ♥ Adrenal glands – Stress response, salt balance
- ♥ Pancreas (Islets of Langerhans) – Regulates blood sugar
- ♥ Gonads (Testes & Ovaries) – Reproduction
- ♥ Pineal gland – Regulates sleep-wake cycle
- ♥ Thymus – Immunity (secretes thymosin)

ENDOCRINE GLANDS IN HUMAN BODY



4 MECHANISM OF HORMONE ACTION



Example :

Insulin (from pancreas) → Blood → Liver / muscles → Uptake of glucose → Lowers blood sugar level.

IMPORTANT NOTE

Hormones are specific in their action. A hormone acts only on those cells that have the specific receptors for it.

Both systems – endocrine and nervous – work together to maintain **homeostasis** and ensure survival.

Stay Focused, Stay Consistent, You'll Succeed



5 HORMONES

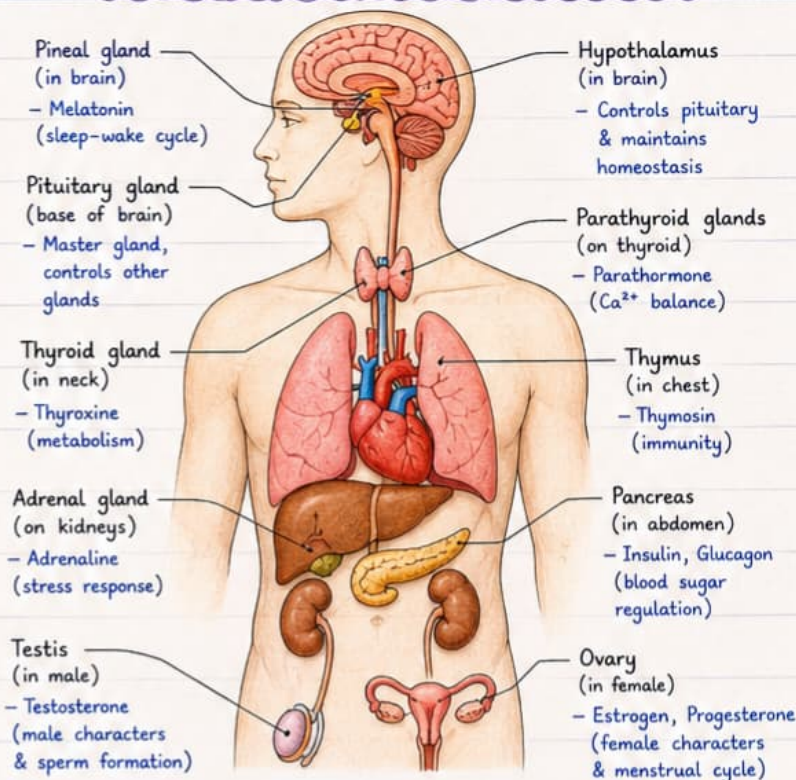
- Hormones are chemical messengers produced in minute amounts by endocrine glands.
- They are transported by blood to reach target organs.
- Effective in very low concentrations.

CHARACTERISTICS OF HORMONES

- Produced in small amounts.
- Released directly into blood.
- Specific in their action.
- Act on target cells/organs.
- Highly potent.
- Short half-life (broken down after action).



MAJOR ENDOCRINE GLANDS OF HUMAN BODY



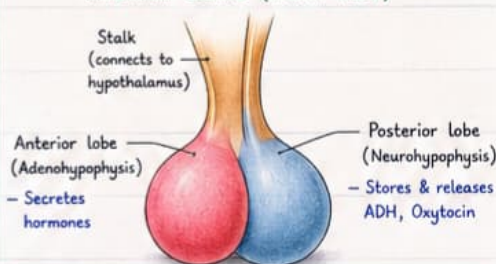
6 MAJOR HORMONES, SOURCE AND FUNCTIONS

HORMONE	SOURCE	MAJOR FUNCTION(S)
Growth hormone (GH)	Pituitary	Growth of bones and muscles
Thyroxine (T_3/T_4)	Thyroid	Regulates metabolism
Parathormone (PTH)	Parathyroid	Regulates Ca^{2+} and phosphate balance
Insulin	Pancreas (β -cells)	Lowers blood glucose level
Glucagon	Pancreas (α -cells)	Raises blood glucose level
Adrenaline	Adrenal medulla	Prepares body for "fight or flight"
Cortisol	Adrenal cortex	Stress response, metabolism, anti-inflammatory
Testosterone	Testis	Male secondary characters, sperm formation
Estrogen	Ovary	Female secondary characters, menstrual cycle
Progesterone	Ovary	Maintains pregnancy
Melatonin	Pineal	Regulates sleep-wake cycle

7 PITUITARY GLAND - THE MASTER GLAND

- Small pea-sized gland at the base of the brain.
- Controls most other endocrine glands.
- Two parts :
 - Anterior pituitary (Adenohypophysis)**
 - Secretes hormones like GH, TSH, ACTH, FSH, LH, Prolactin.
 - Posterior pituitary (Neurohypophysis)**
 - Stores and releases ADH and Oxytocin (made in hypothalamus).

PITUITARY GLAND (FRONT VIEW)



8 HYPOTHALAMUS - THE LINK BETWEEN NERVOUS SYSTEM AND ENDOCRINE SYSTEM

- Located below the thalamus in the brain.
- Maintains homeostasis by regulating :

- | | |
|---|--|
| ☒ Body temperature | ☒ Emotions |
| ☒ Water balance | ☒ Blood pressure |
| ☒ Hunger and thirst | ☒ Controls pituitary gland |
| ☒ Sleep-wake cycle | |

It uses both nerve impulses and hormones to maintain internal balance.



9 CHEMICAL COORDINATION IN ANIMALS

- In simple animals like Hydra, coordination is done by diffusion of chemicals.
- In complex animals, endocrine system works with nervous system for quick and long-term responses.
- Together, they maintain homeostasis and ensure proper functioning of the body.



SUMMARY

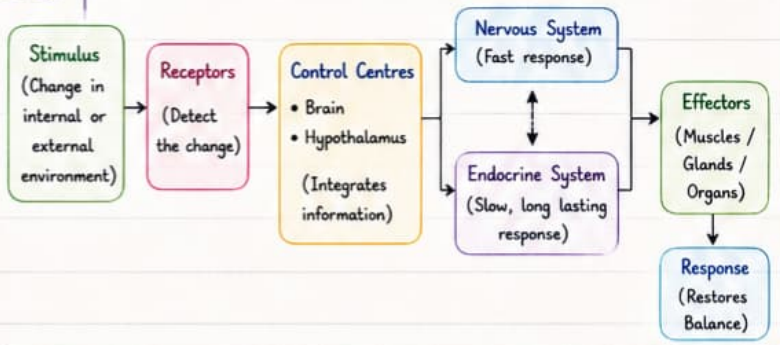
Chemical coordination is achieved by hormones produced in endocrine glands. It works along with nervous system to integrate body functions and maintain stability.

Understand, Revise, Succeed! ♥

10 Nervous System and Chemical Coordination

INTEGRATION OF NERVOUS AND ENDOCRINE SYSTEM

- Nervous system works by electrical impulses - **very fast** but **short lived**.
- Endocrine system works by hormones - **slower** but **long lasting**.
- Both systems are interconnected and **complementary**.



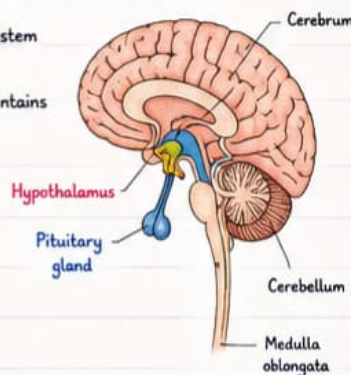
EXAMPLE OF INTEGRATION

During exercise → Brain sends nerve impulses to muscles (fast) to increase heart rate.

At the same time, adrenal gland releases adrenaline (slow, long lasting) to provide more energy. 

11 ROLE OF HYPOTHALAMUS - THE MASTER CONTROL

- Hypothalamus is the link between nervous system and endocrine system.
- It monitors the internal environment and maintains homeostasis.
- Controls pituitary gland (the master gland).
- Regulates:
 - ☒ Body temperature
 - ☒ Water balance (through ADH)
 - ☒ Hunger, thirst & sleep-wake cycle
 - ☒ Emotions and behaviour
 - ☒ Blood pressure and heart rate



12 DISORDERS DUE TO IMBALANCE OF HORMONES

Disorder	Cause (Hormone)	Effect	Example / Condition
Dwarfism	↓ Growth hormone (GH)	Stunted growth	Childhood deficiency of GH
Gigantism	↑ Growth hormone (GH)	Excessive growth	Excess GH in childhood
Acromegaly	↑ Growth hormone (GH)	Enlargement of hands, feet, jaw in adults	Excess GH in adults
Goitre	↓ Thyroxine (T_3/T_4)	Swelling in neck, slow metabolism	Iodine deficiency
Hyperthyroidism	↑ Thyroxine (T_3/T_4)	Weight loss, sweating, irritability, high BMR	Graves' disease
Diabetes mellitus	↓ Insulin	High blood glucose, frequent urination	Insulin deficiency or insulin resistance
Diabetes insipidus	↓ ADH	Excess water loss in urine, excessive thirst	Kidney cannot reabsorb water
Addison's disease	↓ Cortisol	Weakness, low BP, weight loss	Adrenal cortex dysfunction
Cushing's syndrome	↑ Cortisol	Weight gain, high BP, moon face	Excess cortisol secretion

13 FEEDBACK MECHANISM

- Most endocrine functions are regulated by feedback mechanism.

TYPES:

(A) Negative Feedback

Most common type.
Opposes the change and brings the body back to normal.
Example: Regulation of blood glucose by insulin and glucagon.



(B) Positive Feedback

Rare.
Enhances the change until the process is complete.
Example: Childbirth (oxytocin release causes stronger uterine contractions).

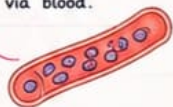


14 SUMMARY OF CHEMICAL COORDINATION

Hormones are chemical messengers produced by endocrine glands.



They act on specific target organs/cells via blood.



Nervous system provides quick responses; endocrine system provides long lasting effects.



Hypothalamus integrates and controls both the systems.



Together they maintain homeostasis and ensure proper growth, development & reproduction. ☒



Remember:

"Balance in chemicals,
Balance in life."

KEY WORDS

Hormone • Endocrine • Homeostasis
Hypothalamus • Pituitary • Target organ

Understand deeply,
Revise smartly,
Succeed surely!

